A Novel WiFi-Based Temporal Precision Detection System With Enhanced CSI Sampling

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***Abstract*—Integrated sensing and communication (ISAC) has emerged as a prominent research focus in the domains of 5G-Advanced (5G-A), 6G, and wireless fidelity (WiFi). As a passive sensing technology, WiFi-based human detection has garnered widespread attention due to its privacy, unobtrusiveness, and convenience in practical applications. This letter introduces a WiFi-based human intrusion detection system utilizing channel state information (CSI). By refining the CSI phase sampling points through the proposed sequence particle filter (SPF) CSI sampling method, the system reduces the amount of data needed for real-time sensing. A comprehensive preprocessing procedure is presented to filter out noise interference and mitigate channel fading during CSI collection. The proposed human intrusion detection algorithm accurately identifies fluctuations in the CSI phase time series, thereby determining the presence of a human and the exact moments of intrusion.**

***Index Terms*—Integrated sensing and communication, channel state information, WiFi, human intrusion detection.**

I. Introduction

ONE OF the main objectives of the sixth-generation (6G) mobile communication is to optimize and improve network environments using wireless signals, thereby enhancing the performance of traditional communication systems [1]. Channel state information (CSI) is crucial as it describes the propagation characteristics of wireless communications and thus senses environmental changes, which is vital for the development of wireless sensor networks. Nowadays, the detection of human presence through the acquisition and processing of CSI data has been extensively studied and utilized.

Widely used human detection methods include those based on dedicated sensors and artificial intelligence (AI) vision. Although these methods can accurately and effectively detect human presence, they are limited to specific application scenarios. Methods based on dedicated sensors require expensive equipment, which must be worn on the body, limiting their widespread practical application. AI vision-based methods require high levels of illumination and can only be used within the line of sight (LoS), thereby posing privacy concerns.

Compared to traditional human detection methods, WiFi-based human sensing leverages integrated sensing and communication (ISAC) capabilities to overcome the aforementioned limitations and achieve the goal of reshaping human activity detection [2], [3]. WiFi-based contactless sensing systems leverage fine-grained CSI data for more accurate human presence detection, without being constrained by LoS conditions.

WiFi-based human activity recognition can utilize either received signal strength indicator (RSSI) or CSI. However, in the case of practical measurement, RSSI is easily affected by temporal and spatial variation, which contributes to most of the estimation errors [4], [5]. Therefore, most existing WiFi-based human recognition systems utilize CSI for sensing [6], [7]. For example, [8] presents a novel object detection system using the quotient of CSI values as the base signal. It distinguishes between three different materials by analyzing their differences through the reconstructed CSI complex.

This letter designs a wireless sensor system for human intrusion detection, the main contributions are as follows:

- The proposed sequence particle filter (SPF) CSI sampling method designs a sampling algorithm to calculate and compare importance weights of CSI phase, selecting the most suitable sampling points for human sensing. This approach refines the dataset by removing unsuitable data and improves detection accuracy by mitigating outliers.
- To address phase interference on certain subcarriers during WiFi CSI collection, this letter proposes a comprehensive preprocessing method. This method preprocesses each subcarrier independently, allowing for detailed observation of detection performance and highlighting phase changes indicative of human presence. Its low complexity also enhances overall detection efficiency.
- The proposed dynamic variance sliding tracking (DVST) algorithm detects human presence by calculating CSI phase changes, allowing for accurate determination of intrusion moments. Its low complexity and independence from extensive pre-existing data enable real-time processing, providing precise timing information on disturbances and enhancing efficiency in practical applications.

Previous works often rely on complex algorithms like deep learning or fingerprint matching for result classification, which depend on extensive pre-training datasets. In contrast, this letter reduces database size, effectively mitigates interference from anomalous sampling points, and features lower complexity in preprocessing and classification, resulting in higher application efficiency.

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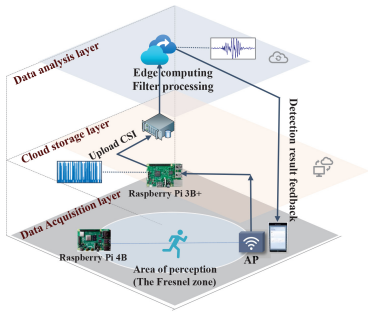

Fig. 1. The simulation diagram of the system model.

II. System Model

The system employs two Raspberry Pi devices with distinct roles: the Raspberry Pi 3B+ monitors and extracts CSI from WiFi signals, while the Raspberry Pi 4B generates traffic to ensure activities are sampled at the same frequency. The router acts as the access point (AP), set to IEEE 802.11ac mode, with an 80MHz bandwidth, a frequency of 5.21GHz, and channel number 157.

Typically, a single CSI time slice is quite large, which imposes significant challenges for real-time processing. As specified by the IEEE 802.11 protocol, the size of the collected CSI data, represented as $CSI_{size} = 32 \times s_{ts} \times r_x \times s_c$, where s_{ts} denotes the number of spatial streams, r_x is the number of receiving antennas, and s_c refers to the number of subcarriers. The size can vary depending on the bandwidth and the number of subcarriers used, ranging from 64 subcarriers at a 20 MHz bandwidth to 128 subcarriers at a 40 MHz bandwidth.

CSI describes the wireless channel by providing each subcarrier's channel frequency response (CFR). Assume that $X(f, t)$, $Y(f, t)$, and $\mathbf{H}(f, t)$ are input signal, output signal, and the CSI matrix. The symbol t in $X(f, t)$, $Y(f, t)$, and $\mathbf{H}(f,t)$ represents the time domain, reflecting the dynamic changes in both the signal and the channel characteristics over time. In this context, $\mathbf{H}(f, t)$ is the channel transmission matrix, which describes the CFR as it varies with time. The relationship is $Y(f, t) = \mathbf{H}(f, t) \times X(f, t) + n$, where n is the channel noise. In the frequency domain, the CSI matrix turns into $\mathbf{H}_{a,b,M_{sub},t}$, where a and b represent transmit and receive antenna, M_{sub} denotes the number of subcarriers. In this experiment, with one transmit and receive antenna, simplifying $\mathbf{H}_{a,b,M_{sub},t}$ into a two-dimensional matrix $\mathbf{H}_{M_{sub},t}$:

$$\mathbf{H}_{M_{sub},t} = \begin{bmatrix} H_{1,1} & H_{1,2} & \cdots & H_{1,t} \\ H_{2,1} & H_{2,2} & \cdots & H_{2,t} \\ \vdots & \vdots & \ddots & \vdots \\ H_{M_{\mathrm{sub},1}} & H_{M_{\mathrm{sub},2}} & \cdots & H_{M_{\mathrm{sub},t}} \end{bmatrix}. \tag{1}$$

In wireless communications, the Fresnel zone is elliptical in shape and serves as an indicator of the detection area. For the F-th Fresnel zone and electromagnetic waves with wavelength $\lambda/2$, the distance of the ellipse can be described as:

$$F\frac{\lambda}{2} = \sqrt{D_1^2 + \overline{r}_n^2} + \sqrt{D_2^2 + \overline{r}_n^2} - (D_1 + D_2), \tag{2}$$

where D_1 and D_2 represent the distances from the ellipse center to the transmitter and receiver, respectively, and $\overline{r}_n$ is the short radius of the ellipse.

III. Algorithms of Human Intrusion Detection

A. Proposed Sequence Particle Filter CSI Sampling Method

In a 128-subcarrier OFDM system, subcarriers are conventionally indexed from -64 to $+63$. This letter converts the indices to range from 0 to 127. After removing the pilot and null subcarriers based on the new indices, the remaining 108 are data subcarriers used for transmitting the original CSI data.

In each sensing cycle, the Raspberry Pi 4B sends N packets, and the SPF CSI sampling method samples from these packets. Each sampled packet contains original complex data for 108 subcarriers. Subsequently, the phase information from the original CSI data in each sampled packet is extracted and recorded in radians:

$$\sigma_i^{1:N} = \left(\sigma_i^1, \sigma_i^2, \ldots, \sigma_i^N\right), \tag{3}$$

where σ_i^N denotes the phase value of the i-th subcarrier within the N-th sampled packet sent during a sensing cycle. The subsequent sampling needs to apply the information from the previous sampling, and the sampled data must follow a joint distribution $\gamma(\sigma_i^{1:N})$. The first sampling follows $\sigma_i^1 \sim \gamma(\sigma_i^1)$, the second sampling follows $\sigma_i^2 \sim \gamma(\sigma_i^2 \mid \sigma_i^1)$, the N-th sampled packet follows $\sigma_i^N \sim \gamma(\sigma_i^N \mid \sigma_i^{1:N-1})$. Through N samplings, the sampled data follows $\sigma_i^{1:N} \sim \gamma(\sigma_i^1)(\sigma_i^2 \mid \sigma_i^1)(\sigma_i^3 \mid \sigma_i^{1:2}) \cdots (\sigma_i^N \mid \sigma_i^{1:N-1})$.

Due to the distribution $\gamma(\sigma_i^{1:N})$ is unknown, assuming its optimal solution in a similar distribution is $q(\sigma_i^{1:N})$. According to the importance sampling theorem, the weight factor is set as $\omega(\sigma_i^{1:N}) = \gamma(\sigma_i^{1:N})/q(\sigma_i^{1:N})$. The relationship between the weight at the N-th sampling and the weight at the $(N-1)$-th sampling is expressed as:

$$\omega\left(\sigma_i^{1:N}\right) = \frac{\gamma\left(\sigma_i^{1:N}\right)}{q\left(\sigma_i^{1:N}\right)}, \tag{4a}$$

$$= \frac{\gamma\left(\sigma_i^{1:N}\right)}{q\left(\sigma_i^N \mid \sigma_i^{1:N-1}\right) q\left(\sigma_i^{1:N-1}\right)} \frac{\gamma\left(\sigma_i^{1:N-1}\right)}{\gamma\left(\sigma_i^{1:N-1}\right)}, \tag{4b}$$

$$= \omega\left(\sigma_i^{1:N-1}\right) \frac{\gamma\left(\sigma_i^{1:N}\right)}{q\left(\sigma_i^N \mid \sigma_i^{1:N-1}\right) \gamma\left(\sigma_i^{1:N-1}\right)}. \tag{4c}$$

Represent the multi-dimensional data $\sigma_i^{1:N}$ as a time series $\sigma_i^{1:K}$ from time 1 to K. $\sigma_i^{1:K}$ follows the distribution $\gamma(\sigma_i^{1:K})$. The distribution $\gamma(\sigma_i^{1:K})$ represents the joint channel state and phase measurement values at all times, i.e., $\gamma(\sigma_i^{1:K}) = p(\sigma_i^{1:K}, z_i^{1:K})$. Substitute this into the equation (4c), the importance weights for the proposed SPF CSI sampling method are represented as:

$$\omega\left(\sigma_i^{1:K}\right) = \omega\left(\sigma_i^{1:K-1}\right) \frac{p\left(\sigma_i^{1:K}, z_i^{1:K}\right)}{q\left(\sigma_i^K \mid \sigma_i^{1:K-1}\right) p\left(\sigma_i^{1:K-1}, z_i^{1:K-1}\right)},$$

$$= \omega\left(\sigma_i^{1:K-1}\right) \frac{p\left(z_i^K \mid \sigma_i^{1:K}, z_i^{1:K-1}\right) p\left(\sigma_i^{1:K}, z_i^{1:K-1}\right)}{q\left(\sigma_i^K \mid \sigma_i^{1:K-1}\right) p\left(\sigma_i^{1:K-1}, z_i^{1:K-1}\right)},$$

$$= \omega\left(\sigma_i^{1:K-1}\right) \frac{p\left(z_i^K \mid \sigma_i^{1:K}, z_i^{1:K-1}\right) p\left(\sigma_i^{1:K-1}, z_i^{1:K-1}\right)}{q\left(\sigma_i^K \mid \sigma_i^{1:K-1}\right) p\left(\sigma_i^{1:K-1}, z_i^{1:K-1}\right)} \times p\left(\sigma_i^K \mid \sigma_i^{1:K-1}, z_i^{1:K-1}\right),$$

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$$= \omega\left(\sigma_i^{1:K-1}\right)\frac{p\left(z_i^K \mid \sigma_i^K\right)p\left(\sigma_i^K, \sigma_i^{K-1}\right)}{q\left(\sigma_i^K \mid \sigma_i^{1:K-1}\right)}. \tag{5}$$

where

$$p\left(z_i^K \mid \sigma_i^K\right) = \frac{1}{\sqrt{2\pi}}\exp\left\{-\frac{\left(z_i^K - \sigma_i^K\right)^2}{2}\right\}, \quad \text{and} \tag{6}$$

$$p\left(\sigma_i^{1:K}, z_i^{1:K}\right) = p\left(z_i^{1:K}\right)\prod_{k=1}^{K} p\left(\sigma_i^k \mid z_i^k\right), \tag{7}$$

represent Gaussian distribution and joint probability distribution. $z_i^{1:K}$ is phase measurement of CSI for all time instances.

Then the weights of the sampling points corresponding to each time instance are obtained. $q(\sigma_i^K)$ denotes the probability of σ_i^K, which is primarily influenced by the preceding state σ_i^{K-1}. Substitute the suboptimal solution $q(\sigma_i^K) = p(\sigma_i^K \mid \sigma_i^{K-1})$ of $q(\sigma_i^K)$ into the importance weight calculation:

$$\begin{aligned}\omega\left(\sigma_i^{1:K}\right) &= \omega\left(\sigma_i^{1:K-1}\right)\frac{p\left(z_i^K \mid \sigma_i^K\right)p\left(\sigma_i^K \mid \sigma_i^{K-1}\right)}{q\left(\sigma_i^K \mid \sigma_i^{1:K-1}\right)},\\ &= \omega\left(\sigma_i^{1:K-1}\right)\frac{p\left(z_i^K \mid \sigma_i^K\right)p\left(\sigma_i^K \mid \sigma_i^{K-1}\right)}{p\left(\sigma_i^K \mid \sigma_i^{K-1}\right)},\\ &= \omega\left(\sigma_i^{1:K-1}\right)p\left(z_i^K \mid \sigma_i^K\right).\end{aligned} \tag{8}$$

During the weight iteration update process, particle degeneracy may occur. The degree of degeneracy is expressed as:

$$\xi_{\text{eff}} = \frac{108}{1 + \text{Var}\left(\text{g}_{*_i}^K\right)}, \quad \text{g}_{*_i}^K = \frac{p\left(\sigma_i^K \mid z_i^{1:K}\right)}{q\left(\sigma_i^K \mid \sigma_i^{K-1}, z_i^K\right)}. \tag{9}$$

When $\xi_{\text{eff}} < \xi_T$, where ξ_T is the threshold for particle degeneracy, resampling is required to avoid particle degeneration issues. In $q(\sigma_i^K \mid \sigma_i^{1:K-1}, z_i^K)$, $\sigma_i^{1:K-1}$ denotes the positions $(\sigma_i^{1:K-1})_j$ of particles after resampling, then:

$$q\left(\sigma_i^K \mid \sigma_i^{1:K-1}, z_i^K\right) = \omega\left(\sigma_i^{1:K-1}\right)q\left[\sigma_i^K \mid \left(\sigma_i^{1:K-1}\right)_j, z_i^K\right]. \tag{10}$$

Combined with $q(\sigma_i^K) = p(\sigma_i^K \mid \sigma_i^{K-1})$, the formula for calculating weights after resampling is expressed as:

$$\begin{aligned}\omega\left(\sigma_i^{1:K}\right) &= \omega\left(\sigma_i^{1:K-1}\right)\frac{p\left(z_i^K \mid \sigma_i^K\right)p\left(\sigma_i^K \mid \sigma_i^{K-1}\right)}{q\left(\sigma_i^K \mid \sigma_i^{1:K-1}\right)},\\ &= \omega\left(\sigma_i^{1:K-1}\right)\frac{p\left(z_i^K \mid \sigma_i^K\right)p\left(\sigma_i^K \mid \sigma_i^{K-1}\right)}{\omega\left(\sigma_i^{1:K-1}\right)q\left[\sigma_i^K \mid \left(\sigma_i^{1:K-1}\right)_j, z_i^K\right]},\\ &= \frac{p\left(z_i^K \mid \sigma_i^K\right)p\left(\sigma_i^K \mid \sigma_i^{K-1}\right)}{q\left[\sigma_i^K \mid \left(\sigma_i^{1:K-1}\right)_j, z_i^K\right]} = p\left[z_i^K \mid \left(\sigma_i^K\right)_j\right].\end{aligned} \tag{11}$$

According to particle weights obtained after iteration, select l CSI phase values with the highest weights from N sampling points, denoted by symbol $\hat{\sigma}_i^L$, where $L = 1, 2, \ldots, l$.

B. Proposed Preprocessing Method for CSI Data

The sampled CSI phase values $\hat{\sigma}_i^L$ require preprocessing to filter out noise and mitigate channel fading. The phase unwrapping can be expressed as:

$$\Xi_i^L = \begin{cases}\hat{\sigma}_i^L - 2\pi, \text{ if } \ \hat{\sigma}_i^L - \hat{\sigma}_{i-1}^L > \pi,\\ \hat{\sigma}_i^L + 2\pi, \text{ if } \ \hat{\sigma}_{i-1}^L - \hat{\sigma}_i^L > \pi,\\ \hat{\sigma}_i^L, \qquad \text{if } \left|\hat{\sigma}_i^L - \hat{\sigma}_{i-1}^L\right| < \pi,\end{cases} \tag{12}$$

where Ξ_i^L represents the CSI phase value that has already been unwrapped. The original CSI phase value $\hat{\sigma}_0^L$ corresponding to the subcarrier with index 0 remains unchanged, as $\Xi_0^L = \hat{\sigma}_0^L$.

After unwrapping CSI phase for 108 subcarriers in a sampled data packet, the linear transformation is applied to correct phase offsets. The first step is to determine the slope as $\eta^L = (\Xi_{107}^L - \Xi_0^L)/108$ and offset as $\upsilon^L = (\sum_{i=0}^{107}\Xi_i^L)/108$. Linear transformation is expressed as:

$$\psi_i^L = \begin{cases}\Xi_i^L - \eta^L \times \left(i - \frac{108}{2}\right) - \upsilon^L, & \text{if } \ 0 \le i < 54,\\ \Xi_i^L - \eta^L \times \left(i - \frac{108}{2} + 1\right) - \upsilon^L, & \text{if } \ 54 \le i \le 107,\end{cases} \tag{13}$$

where ψ_i^l denotes phase values after the linear transformation.

Next, CSI phase values undergo the proposed enhanced median absolute deviation (EMAD) filter. The median of each window of the filter can be expressed as:

$$\tilde{h}_i^L = \begin{cases}\text{if } \ 1 \le L < C+1:\\ \text{median}\left(\psi_i^1, \psi_i^2, \ldots, \psi_i^{L+C}\right),\\ \text{if } \ C+1 \le L < l-C+1:\\ \text{median}\left(\begin{array}{l}\psi_i^{L-C}, \psi_i^{L-C+1}, \ldots, \psi_i^{L-1},\\ \psi_i^L, \psi_i^{L+1}, \ldots, \psi_i^{L+C}\end{array}\right),\\ \text{if } \ l-C+1 \le L \le l:\\ \text{median}\left(\psi_i^l, \psi_i^{l-1}, \ldots, \psi_i^{L-C}\right),\end{cases} \tag{14}$$

where C is the window length of the EMAD filter. Then, compute the median absolute deviations for each window as:

$$\tilde{d}_i^L = \begin{cases}\text{if } \ 1 \le L < C+1:\\ \text{median}\left(\left|\psi_i^1 - \tilde{h}_i^L\right|, \left|\psi_i^2 - \tilde{h}_i^L\right|, \ldots, \left|\psi_i^{L+C} - \tilde{h}_i^L\right|\right),\\ \text{if } \ C+1 \le L < l-C+1:\\ \text{median}\left(\begin{array}{l}\left|\psi_i^{L-C} - \tilde{h}_i^L\right|, \left|\psi_i^{L-C+1} - \tilde{h}_i^L\right|, \ldots,\\ \left|\psi_i^{L-1} - \tilde{h}_i^L\right|, \left|\psi_i^L - \tilde{h}_i^L\right|, \left|\psi_i^{L+1} - \tilde{h}_i^L\right|,\\ \ldots, \left|\psi_i^{L+C} - \tilde{h}_i^L\right|\end{array}\right),\\ \text{if } \ l-C+1 \le L \le l:\\ \text{median}\left(\begin{array}{l}\left|\psi_i^l - \tilde{h}_i^L\right|, \left|\psi_i^{l-1} - \tilde{h}_i^L\right|, \ldots,\\ \left|\psi_i^{L-C} - \tilde{h}_i^L\right|\end{array}\right).\end{cases} \tag{15}$$

The threshold for each EMAD filter window is $\tau_i^L = \alpha \times n \times \tilde{d}_i^L$, where α is the scaling factor and n is the multiple of the median absolute deviation for outlier detection. The subsequent step is to complete the EMAD filter as follows:

$$\varphi_i^L = \begin{cases}\tilde{h}_i^L, \text{ if } \left|\psi_i^L - \tilde{h}_i^L\right| \ge \tau_i^L,\\ \psi_i^L, \text{ if } \left|\psi_i^L - \tilde{h}_i^L\right| < \tau_i^L,\end{cases} \tag{16}$$

where φ_i^L denotes the phase value after EMAD filtering.

Next, the phase values are processed through the proposed adaptive polynomial dynamic fit (APDF) filter. The first step is to determine the APDF filter coefficients based on the filter window length S and the polynomial order ϱ. Then an array $R \in \mathbb{C}^{1\times S}$ is defined based on the window length s as follows:

$$\begin{aligned}R = \Bigg[&r_0 = -\frac{S-1}{2}, r_1 = -\frac{S-1}{2} + 1, r_2 = -\frac{S-1}{2} + 2,\\ &\ldots, r_{S-2} = \frac{S-1}{2} - 1, r_{S-1} = \frac{S-1}{2}\Bigg].\end{aligned} \tag{17}$$

Afterward, based on the elements inside the array R, a Vandermonde matrix $A \in \mathbb{C}^{S\times(\varrho+1)}$ can be constructed:

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$$A = \begin{bmatrix} r_0^0 & r_0^1 & r_0^2 & \cdots & r_0^{\varrho} \\ r_1^0 & r_1^1 & r_1^2 & \cdots & r_1^{\varrho} \\ r_2^0 & r_2^1 & r_2^2 & \cdots & r_2^{\varrho} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ r_{S-1}^0 & r_{S-1}^1 & r_{S-1}^2 & \cdots & r_{S-1}^{\varrho} \end{bmatrix}, \tag{18}$$

where r_{S-1}^{ϱ} denotes the ϱ power of element r_{S-1} in array R.

The polynomial coefficients are then solved using the least squares method. The steps are as follows: to represent unit weights, a unit matrix $I \in \mathbb{C}^{S\times S}$ is first constructed:

$$I = \begin{bmatrix} I_{11}=1 & I_{12}=0 & I_{13}=0 & \cdots & I_{1S}=0 \\ I_{21}=0 & I_{22}=1 & I_{23}=0 & \cdots & I_{2S}=0 \\ I_{31}=0 & I_{32}=0 & I_{33}=1 & \cdots & I_{3S}=0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ I_{S1}=0 & I_{S2}=0 & I_{S3}=0 & \cdots & I_{SS}=1 \end{bmatrix}. \tag{19}$$

The goal of the least squares method is to find a coefficient matrix that minimizes the sum of squared residuals between matrices A and I. This minimization problem is expressed as:

$$Q = \arg\min_{Q} ||AQ - I||^2, \tag{20}$$

where Q is the coefficient matrix to be determined. The minimization problem can be solved optimally by solving the normal equation. The normal equation is expressed as $A^T AX = A^T I$, where the solution X is the coefficient matrix Q. The coefficient matrix $Q \in \mathbb{C}^{(\varrho+1)\times S}$ can be expressed as:

$$Q = \begin{bmatrix} Q_{11} & Q_{12} & Q_{13} & \cdots & Q_{1S} \\ Q_{21} & Q_{22} & Q_{23} & \cdots & Q_{2S} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ Q_{(\varrho+1)1} & Q_{(\varrho+1)2} & Q_{(\varrho+1)3} & \cdots & Q_{(\varrho+1)S} \end{bmatrix}. \tag{21}$$

The elements of the first row of matrix Q are selected to form the APDF filter coefficient vector, which is expressed as:

$$E = \Big[E_1 = Q_{11}, E_2 = Q_{12}, E_3 = Q_{13}, \ldots, E_S = Q_{1S}\Big]. \tag{22}$$

Finally, CSI phase values are convolved with the coefficient vector E, expressed by the following convolution formula:

$$\delta_i^{\zeta} = \sum_{m=1}^{S} E_m \cdot \varphi_i^{\zeta-m+1}, \quad \zeta = 1, 2, \ldots, l+S-1, \tag{23}$$

where δ_i^{ζ} represents the phase values sequence after convolution, specified as: $\delta_i^{\zeta} = [\delta_i^1, \delta_i^2, \ldots, \delta_i^{l+S-1}]$. To maintain system consistency, the convolution output is truncated, with the start and end indices of the truncated sequence as follows:

$$\text{start} = \lfloor \frac{S}{2} \rfloor, \quad \text{end} = \lfloor \frac{S}{2} \rfloor + l - 1. \tag{24}$$

The truncated sequence of CSI phase values is shown below:

$$\theta_i^L = \left[\theta_i^1 = \delta_i^{\lfloor \frac{S}{2} \rfloor}, \theta_i^2 = \delta_i^{\lfloor \frac{S}{2} \rfloor + 1}, \ldots, \theta_i^l = \delta_i^{\lfloor \frac{S}{2} \rfloor + l - 1}\right]. \tag{25}$$

With the preprocessed phase values θ_i^L obtained, the next step is to implement the human intrusion detection algorithm.

C. Proposed Dynamic Variance Sliding Tracking Algorithm

This letter proposes the DVST algorithm for CSI phase values. The first step is to set a window of length U, which traverses the entire CSI phase value sequence θ_i^L. The mean and variance of phase values within the window are calculated after each slide. Mean values and variances are expressed as:

$$\mu_i^{\varpi} = \frac{1}{U} \times \left(\sum_{\Gamma=\varpi}^{\varpi+U-1} \theta_i^{\Gamma}\right), \varpi = 1, 2, \ldots, l-U+1, \tag{26}$$

$$\hat{v}_i^{\varpi} = \frac{1}{U} \times \left[\sum_{\Gamma=\varpi}^{\varpi+U-1} \left(\theta_i^{\Gamma} - \mu_i^{\varpi}\right)^2\right], \varpi = 1, 2, \ldots, l-U+1. \tag{27}$$

The next step is to define a threshold ρ to detect abnormal phase fluctuations from human intrusion, expressed as:

$$\rho = \hat{\rho} \times \text{median}\Big(\hat{v}_i^1, \hat{v}_i^2, \ldots, \hat{v}_i^{l-U+1}\Big), \tag{28}$$

where $\hat{\rho}$ denotes the expansion factor. Environmental interference like noise can cause significant phase value fluctuations. A few variances exceeding the threshold can not confirm human intrusion. Human intrusion is confirmed only if a certain number G of consecutive variances exceed the threshold:

$$\begin{cases} 1, \text{ if } \hat{v}_i^{\chi} > \rho \text{ for all } \chi = \beta, \beta+1, \ldots, \beta+G-1, \\ 0, \text{ if } \hat{v}_i^{\chi} > \rho \text{ not for all } \chi = \beta, \beta+1, \ldots, \beta+G-1, \end{cases} \tag{29}$$

where $\beta = 1, 2, \ldots, l-U-G+2$. Output 1 indicates the presence of human intrusion, and output 0 indicates the absence of human intrusion. If all outputs within a sensing cycle are 0, it indicates no human intrusion during that cycle.

For the single input multiple output (SIMO) system with a single transmitting antenna and $\eth$ receiving antennas, $\mathbf{h}_{\eth}$ represents the channel state from the transmitting antenna to the $\eth$-th receiving antenna. The CSI particle set is defined as $\{\mathbf{h}_{\eth}(\sigma_i^{1:N}), \omega_{\eth}(\sigma_i^{1:N})\}$, representing N CSI particles and their corresponding weights. Using the proposed SPF CSI sampling method, l sampling points are selected from the N CSI particles, represented as $\{\mathbf{h}_{\eth}, \hat{\sigma} i^L\}$. The preprocessing stage begins with applying the phase unwrapping method, yielding the unwrapped CSI phase values, represented as $\{\mathbf{h}_{\eth}, \Xi_i^L\}$. This is followed by a linear transformation process, resulting in $\{\mathbf{h}_{\eth}, \psi_i^L\}$. Subsequently, the data is processed through the EMAD filter, producing $\{\mathbf{h}_{\eth}, \varphi_i^L\}$. Finally, the APDF filter is applied, yielding $\{\mathbf{h}_{\eth}, \theta_i^L\}$. After preprocessing, the DVST algorithm can be applied for human intrusion detection.

IV. Experimental Results

This section presents experimental results demonstrating the effectiveness of the human intrusion detection system in real scenarios. The experiment takes place in a 5 × 4.5 meter indoor laboratory, with the router and Raspberry Pi 4B positioned 2.7 meters apart. The system's sensing period is set to 10 seconds. Fig. 2 compares two scenarios: no human intrusion and human intrusion during a sensing cycle. The first subfigure presents the raw CSI phase values within a sensing period, where both waveforms appear chaotic, making it challenging to discern human intrusion. The second subfigure presents phase values after applying the SPF sampling method and data preprocessing. It reveals a stable phase waveform without human intrusion and noticeable fluctuations when intrusion occurs. The third subfigure depicts the waveform of the phase variance array $\hat{v}_i^{\varpi}$, which remains stable without human intrusion but increases at specific time segments during intrusion. The proposed DVST algorithm identifies the intrusion onset at approximately 5.795 seconds into the sensing period.

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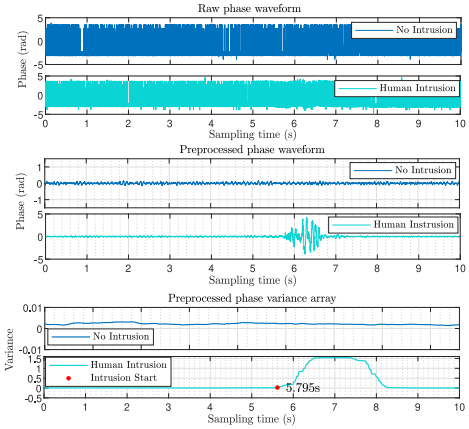

Fig. 2. Comparison of CSI phase value waveforms before and after preprocessing.

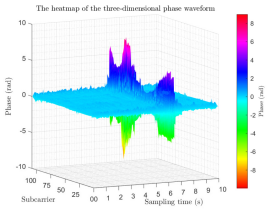

Fig. 3. Waveforms of CSI phase values when the human intrusion occurs.

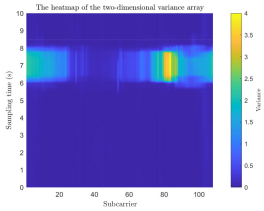

Fig. 4. Heatmap of the variance arrays when the human intrusion occurs.

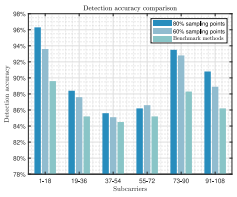

Fig. 5. Detection accuracy comparison.

Fig. 3 displays the waveforms of CSI phase values carried by 108 subcarriers. It depicts the scenario where human intrusion occurs, showing varying fluctuations in the CSI phase waveforms during the same time segments. Fig. 4 provides a more intuitive representation of the sensitivity of each data subcarrier to detecting the human intrusion. It presents the occurrence of human intrusion, showing an increase in variance values between the 6th and 8th seconds, indicating the detection of human intrusion during this period. Fig. 5 presents a comparison of detection accuracy. Unlike benchmark methods, which process all collected CSI data points directly, the method proposed in this letter filters the sampling points based on their importance, using only the top 80% and 60% for detection and accuracy comparison.

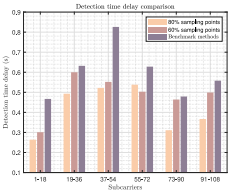

Fig. 6. Detection time delay comparison.

To ensure the accuracy of the results, the detection accuracy for 108 subcarriers is analyzed separately. As shown in the figure, the accuracy of the proposed method remains higher than that of the benchmark methods across all groups of subcarriers, for both 80% and 60% sampling points. Fig. 6 presents the comparison of detection time delay. The average detection time delay is calculated for each group of subcarriers. The results indicate that the method proposed in this letter, even when using only 80% and 60% of the sampling points, exhibits consistently lower time delay, significantly reducing detection latency compared to the benchmark methods.

V. Conclusion

In this letter, a WiFi-based human intrusion detection system utilizing CSI was proposed. The SPF CSI sampling method selected optimal CSI sampling points based on particle weights, reducing the required data for processing and enhancing real-time sensing capabilities. The CSI data preprocessing method mitigated channel fading interference and deviations, improving sensing accuracy. The DVST algorithm effectively identified CSI phase fluctuations due to human intrusion, accurately determining the exact moments of these fluctuations. In future developments, ensuring the confidentiality and anti-eavesdropping of sensed information is a critical challenge for optimizing ISAC systems.

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